

NimbusCompute Forecast

Assumptions, Limitations, Interpretation and Quality Assessment

Forecast period: June-August 2026 | Forecast cut-off: 6 June 2026

This document describes the forecast exactly as implemented in the current repository. It does not change the model or recommend code changes. Its purpose is to make every material assumption, judgement call, ambiguity and quality limitation explicit.

1. Purpose and intended use

The model produces a three-month scenario forecast for two related but different financial views:

- Consumption revenue: the value of cloud resources expected to be consumed in each service month.
- Cash collections: the cash expected to be received in each calendar month.

The model is designed as a transparent management-planning tool for Finance and Sales leadership. It is not a statutory revenue-recognition model, a probability-of-default model, or a fully automated production forecasting system.

The primary value of the model is explainability. Each result can be traced to current customer usage, recent growth coefficients, contractual price and discount assumptions, open receivables, historical payment timing, and explicitly selected CRM opportunities.

1.1 Forecast outputs

View	Downside	Base	Upside
Consumption revenue, Jun-Aug 2026	Lower organic trajectory and limited pipeline	Recent organic trajectory plus selected pipeline	Higher organic trajectory plus full upside pipeline
Cash collections, Jun-Aug 2026	Same existing AR, lower future usage, limited prepayments	Same existing AR, base future usage, selected prepayments	Same existing AR, higher future usage, full upside prepayments

1.2 Key interpretation rule

Consumption revenue and cash collections must not be interpreted as the same metric. A customer can consume services in one month and pay in a later month, while annual-prepayment opportunities can create a large cash receipt before the related consumption is recognised over time.

2. Data scope and source hierarchy

The model combines billing, usage, contract, invoice and CRM data. Where sources conflict, the model uses different sources for different purposes rather than assuming that one table is universally authoritative.

Source	Primary use in the model	Important limitation
monthly_usage	Historical product consumption and May forecast anchor	Contains operational usage, but not future commercial commitments.
price_list	List-price lookup by product and effective date	Eligibility rules for some price changes are not fully encoded at customer-renewal level.
contracts	Discounts, billing frequency, term and prepayment structure	Contract status, end date and auto-renewal can produce ambiguous future treatment.
billing_customers	Primary current-customer status filter	Current status can conflict with recent usage activity.
invoices	Open AR, payment dates and historical payment timing	Paid-invoice timing is not the same as probability of eventual collection.
crm_opportunities	Incremental consumption and prepayment scenarios	Overlapping opportunities and unresolved account mapping require judgement.
business_events	Context for migrations, constraints, pricing and system changes	Events provide qualitative context, not complete transaction-level correction rules.

2.1 Forecast cut-off

The forecast is constructed as of 6 June 2026. Information that became observable after that date should not be used to reconstruct the state of receivables or to assess whether a pipeline deal was already realised. The invoice logic therefore reconstructs open AR point-in-time rather than relying only on the final payment status stored in the extract.

3. Consumption revenue methodology

3.1 Usage normalisation

Usage is aggregated at customer, product and month level. Each product is mapped to its economically relevant native quantity:

- `compute_standard` -> `compute_hours`
- `gpu_a100` and `gpu_accelerated` -> `gpu_hours`
- `network_egress` -> `egress_tb`
- `object_storage` -> `storage_tb_avg`

The legacy GPU SKU and current GPU SKU are combined for analytical interpretation because the business-event data describes a product migration rather than the launch of an economically separate product. Pricing is still applied using the source SKU before the products are grouped into the common GPU family.

3.2 Active-customer cohort

The organic forecast includes only customers currently marked as active in `billing_customers`. This is an explicit planning assumption: the future recurring base should represent the customers considered active at the cut-off date.

This creates survivorship bias in historical charts derived from the same filtered dataset. Historical values in the forecast notebook should therefore be read as the historical performance of today's active cohort, not as total historical company revenue.

Impact on forecast quality: this improves relevance for forecasting the retained base, but understates historical churn and can make the cohort's past growth look stronger or smoother than the company's full historical performance.

3.3 Forecast anchor

May 2026 is the latest complete usage month and is used as the anchor for every future customer-product quantity. The model does not forecast directly from an average of several months; recent March-April-May dynamics determine the growth coefficient, while May determines the starting level.

Impact: the model reacts quickly to recent customer behaviour, but any one-off spike, temporary capacity issue, delayed workload or measurement anomaly in May directly affects the entire three-month forecast.

3.4 Growth-coefficient hierarchy

Selected large customers are modelled using customer-product coefficients. All remaining customers use product-level coefficients. This hybrid approach reflects concentration: a small number of customers materially influence total consumption and should not be diluted into a portfolio average.

- CUST-0006 and CUST-0018 receive individual customer-product coefficients.
- The remaining active customer base receives product-level coefficients calculated after excluding those large customers.

Impact: customer-specific modelling improves sensitivity to major accounts, while product-level pooling reduces noise among smaller accounts. However, only the explicitly selected large customers receive bespoke treatment; other concentrated or unusual accounts remain inside product averages.

3.5 Coefficient calculation

For March, April and May, the model calculates two multiplicative movements: April/March and May/April. If the previous-month quantity is zero, the corresponding coefficient is set to 1.0. This avoids division by zero and treats the unobservable growth rate as neutral.

The scenario coefficients are then bounded:

Scenario	Raw basis	Applied range	Interpretation
Downside	Lower of the two observed coefficients	0.90 to 1.00	At most a 10% monthly decline and no positive growth.
Base	Geometric mean of the two observed coefficients	0.95 to 1.05	Compounded recent trend, limited to +/-5% per month.
Upside	Higher of the two observed coefficients	1.00 to 1.10	No decline and at most 10% monthly growth.

The use of caps is a deliberate judgement call. It prevents a small denominator or short-lived workload ramp from producing an explosive three-month extrapolation. The trade-off is that genuine rapid growth above the cap is intentionally suppressed.

3.6 Compounding

Each scenario coefficient is compounded from the May anchor: one period for June, two for July and three for August. This assumes the same monthly movement persists throughout the forecast horizon.

Impact: compounding is appropriate for multiplicative usage growth, but uncertainty grows with each month. August is structurally less reliable than June because the coefficient is applied three times.

4. Pricing and contract assumptions

4.1 Price-list treatment

Revenue is calculated as forecast quantity multiplied by the selected list price and by one minus the applicable discount.

Known price changes are represented explicitly for `compute_standard` and `object_storage`. The object-storage price change is treated as universally applicable from its effective date.

The July 2025 compute price change is applied to all compute usage from the effective date in the implemented SQL. The business-event description states that the price change applies to new and renewing contracts, but the current model does not reconstruct customer-level renewal eligibility. This is therefore a broad-pricing approximation.

Expected directional impact: applying the lower compute price broadly can understate consumption revenue for customers that were still on older terms. In the current dataset, the estimated effect is small relative to total revenue, so it does not alter the scenario narrative.

4.2 Discounts and contract periods

Contract discounts are applied when the forecast month falls within the modelled contract period. Active contracts with auto-renewal are extended beyond the forecast horizon. Non-renewing or non-active contracts stop at the recorded contract end month.

Once a contract is considered expired, the model applies no contractual discount and therefore falls back to list price. This implicitly assumes that the customer continues consuming but no replacement commercial terms are available.

Impact: this can overstate revenue if an expired customer would renew at a similar or deeper discount, or understate revenue if the customer would churn rather than continue at list price.

4.3 Contract overlap risk

The implemented logic joins usage to all qualifying contract rows. The modelling assumption is that each customer-month has one relevant contract. If contract periods overlap, usage and revenue can be duplicated. The current results rely on the dataset not producing material overlaps after the applied exclusions and auto-renewal treatment.

Quality implication: this is primarily a data-integrity risk rather than a forecasting assumption. A duplicate contract match would bias all three scenarios upward for the affected customer-month.

4.4 Explicit exclusion of CON-1888

CON-1888 is excluded from the organic contract logic because the associated commercial event is treated separately through the CRM pipeline. This is intended to prevent a prepayment or discount structure from being counted both as existing organic business and as a new opportunity.

The link between the contract and the opportunity is not fully resolved across CRM and billing identifiers. The exclusion is therefore a business-mapping assumption, not a formally proven key relationship.

5. Special customer treatment

5.1 Large customers

Large customers are separated because the exploratory analysis shows that usage is concentrated. A portfolio-level coefficient alone would hide material account-level ramps and reversals.

5.2 CUST-0038 and CUST-0047 in Downside

The current Downside scenario sets the future organic contribution of CUST-0038 and CUST-0047 to zero because of uncertainty around their commercial status. Base and Upside retain their forecast usage.

This is a deliberately severe downside assumption. It removes the full current consumption of those customers, not merely the incremental value of an uncertain renewal or expansion opportunity.

Expected directional impact: Downside consumption and downstream cash are lower than they would be under a scenario that preserved the existing run rate but excluded only incremental growth. The assumption increases scenario separation and should be interpreted as a potential full-loss case.

6. CRM opportunity methodology

Pipeline opportunities are treated as incremental to the May organic customer base. The model uses scenario selection rather than applying the CRM probability field uniformly to every opportunity.

6.1 Scenario selection

Scenario	Included opportunity treatment	Meaning
Downside	OPP-1039 only	Only the most certain selected new deal contributes.
Base	OPP-1039 plus OPP-1036	The selected Vertex compute component materialises.
Upside	OPP-1039 plus OPP-1037	The full Vertex umbrella opportunity materialises.

6.2 Vertex overlap and ambiguity

The Vertex-related opportunities overlap. OPP-1035 and OPP-1036 appear to be component opportunities, while OPP-1037 is described as an umbrella saving plan. The product components sum to approximately 640K of monthly consumption, while the umbrella opportunity is 690K. Because the relationship is not a clean arithmetic sum, the model does not add all three opportunities together.

The current scenario interpretation is:

- Base: only the selected compute component OPP-1036 is realised.
- Upside: the full umbrella opportunity OPP-1037 is realised.
- OPP-1035 is not added separately in either scenario.

This is a management-scenario assumption rather than a direct consequence of stage or probability fields. All three opportunities are marked as Commit and have high sales probability, so the scenario distinction depends on interpretation of deal structure.

Quality implication: Vertex is the dominant source of scenario dispersion, particularly in July cash. Any misunderstanding of whether the opportunities are alternatives, components or sequential deal versions can materially change both consumption and cash.

6.3 Probability field

CRM probability is not used as a universal mathematical multiplier in the current scenario logic. Instead, uncertainty is represented through discrete scenario inclusion. Therefore Base should not be interpreted as a probability-weighted expected value across all pipeline outcomes.

Impact: the scenarios are easier to explain operationally, but the Base case is a selected commercial state rather than a statistically calibrated expected value.

6.4 Start month and ramp

Selected opportunity consumption begins from the expected start month and is added as a recurring monthly amount. The current scenario logic does not apply a detailed customer ramp beyond the opportunity values used in the notebook.

Impact: the forecast is sensitive to start-date slippage. A one-month delay in a large opportunity can materially shift both consumption and cash between July and August or outside the forecast horizon.

7. Cash collections methodology

Cash is modelled independently from consumption and consists of three components:

1. Existing open accounts receivable as of the cut-off date.
2. Collections generated by future postpaid organic usage.
3. Annual prepayments from selected CRM opportunities.

7.1 Point-in-time AR reconstruction

An invoice is treated as open at 6 June 2026 when it was issued by that date and was either unpaid or paid only after that date. This prevents look-ahead bias from using final payment status to infer what Finance knew at the cut-off.

Usage invoices are retained, while void-pending and non-usage records are excluded by the implemented filters. The very large test invoice is therefore excluded from the forecast. The code does not rely solely on a dedicated test flag; exclusion results from invoice-type and status logic.

7.2 Near-term AR

Open receivables whose due-date-plus-seven-days timing falls inside the forecast window are scheduled into the corresponding month at 100% of face value.

Assumption: non-severely-overdue receivables are fully collectible and the expected timing is approximately seven days after due date.

Impact: this can overstate cash if current invoices experience default, dispute or slippage. Conversely, it can shift cash too late for customers that routinely pay before due date.

7.3 Overdue AR recovery

Older overdue balances are assigned recovery percentages by aging bucket. The implemented rates are judgemental rather than statistically estimated from resolved write-off outcomes.

Aging bucket	Recovery assumption	Interpretation
0-30 days overdue	80%	Most recent overdue balance remains largely collectible.
31-90 days overdue	50%	Meaningful deterioration in collectability.
91-180 days overdue	20%	Only a minority is expected to recover.
180+ days overdue	5%	Very old AR is treated as largely impaired.

Expected recovery is distributed evenly across June, July and August. This smooths timing because invoice-specific collection plans are unavailable.

Impact: the total recovery amount may be reasonable as a planning haircut, but the monthly timing is synthetic. Actual collections may be concentrated in one month or may not occur within the forecast period.

7.4 Existing AR held constant across scenarios

Existing AR and overdue recovery are identical in Downside, Base and Upside. Scenario differences are intended to isolate commercial uncertainty in future usage and opportunities rather than collections risk on the existing balance.

Impact: the Downside is not a full liquidity stress case. It does not combine weak consumption with lower AR recovery or slower payment timing.

7.5 Historical payment-lag distribution

Future postpaid consumption is converted into cash using the historical distribution of payment timing among paid usage invoices. Lag is measured from service month to paid month, and paid amounts are normalised into weights.

Critical interpretation: this is a timing model conditional on payment. It is not a model of whether a future invoice will ultimately be collected. The weights sum to approximately 100%, so the method assumes full eventual collection of newly generated postpaid invoices.

Impact: this can overstate lifetime collectible cash when bad debt exists. Within the three-month horizon, part of the assumed collection falls outside the window, so the effect on reported June-August cash is moderated but not eliminated.

7.6 Population mismatch in lag estimation

The historical lag is estimated from all paid usage invoices, while the future-use application excludes annual-prepay customers. As a result, the training population and application population are not perfectly aligned.

Expected impact: moderate timing bias. If annual-prepay-labelled customers have systematically different payment dates, the global lag can shift cash between July, August and later months.

7.7 Billing-frequency selection

Future cash treatment uses the latest active contract's billing frequency for each customer. Annual-prepay customers are excluded from future postpaid cash to avoid double counting prepaid economics.

The implemented selection does not consistently re-evaluate whether a non-renewing annual-prepay contract has expired during the forecast period. Therefore a customer can be treated as annual prepay for cash even when the consumption model has stopped applying that contract's discount.

Expected directional impact: future cash can be understated for customers whose expired annual-prepay contract would otherwise be treated as postpaid or pay-as-you-go. This is not material enough to change the overall narrative but reduces internal consistency between consumption and cash logic.

7.8 New-opportunity prepayments

Annual prepayments are calculated from committed spend and contract term. For a 24-month commitment, the model uses 12/24 of committed spend as the first-year prepayment and assigns it to the expected payment/start month.

These amounts are scenario-specific and can dominate monthly cash. In particular, the full Vertex umbrella prepayment is the main reason Upside July cash is substantially higher than Base.

Impact: cash is highly concentrated in a small number of discrete deal outcomes. The monthly cash forecast should therefore be presented with a bridge between recurring collections and conditional prepayments rather than as a smooth operating run rate.

8. Scenario outcomes and interpretation

Metric (K USD)	June	July	August
Downside consumption	1,551	1,529	1,509
Base consumption	1,631	1,898	1,937
Upside consumption	1,673	2,446	2,535
Downside cash	2,986	1,463	1,201
Base cash	2,986	3,773	1,254
Upside cash	2,986	8,379	1,286

8.1 June

June cash is similar across scenarios because it is driven mainly by existing AR and the selected OPP-1039 prepayment, while scenario-specific Vertex cash begins later. Consumption scenarios already differ because of organic coefficients and customer exclusions, but the cash lag delays much of the effect.

8.2 July

July is the most scenario-sensitive month. Base includes the selected Vertex compute component, while Upside includes the full umbrella deal and its annual prepayment. The resulting cash difference should be interpreted as commercial-deal concentration, not as ordinary month-over-month operating growth.

8.3 August

August consumption reflects three periods of compounded organic assumptions and therefore carries the greatest organic-model uncertainty. Cash is much lower than July because the major annual prepayments are concentrated earlier and most August receipts come from AR and lagged postpaid usage.

9. Assumption and uncertainty register

Assumption / issue	Direction of possible bias	Potential materiality	Main affected output
May used as the sole starting level	Either direction	Medium	All organic consumption
Current active cohort used for history	Historical trend may look stronger	Medium	Interpretation and coefficient context
Growth caps	Suppresses extreme upside and downside	Medium	Organic scenarios
Zero-denominator fallback = 1.0	Neutralises new-product growth	Low-Medium	Specific customer-product rows
Compute price applied broadly after Jul-2025	Likely revenue understatement	Low	Consumption revenue
Expired contract falls back to no discount	Either direction	Medium for affected customers	Consumption revenue
Potential contract overlap	Upward if duplicate matches occur	Potentially high but data-dependent	Consumption and cash
CUST-0038/0047 set to zero in Downside	Makes Downside lower	Medium	Downside consumption and cash
Vertex scenario interpretation	Either direction	High	July consumption and cash
CRM probability not mathematically weighted	Base is not expected value	High interpretive impact	Pipeline forecast
Near-term AR collected at 100%	Possible cash overstatement	Medium	June-July cash
Judgemental overdue recovery rates	Either direction	Medium	All cash months
Recovery spread evenly across months	Timing bias	Low-Medium	Monthly cash profile
Lag based only on paid invoices	Overstates eventual collection	Medium	Future usage cash
Lag training/application populations differ	Timing bias	Low-Medium	July-August cash
Latest active billing frequency used	May understate postpaid cash	Low-Medium	Future usage cash
Existing AR constant across scenarios	Downside not a liquidity stress	High interpretive impact	Scenario comparison
Large prepayments treated discretely	High month concentration	High	July cash

10. Reproducibility and artifact consistency

The core numerical results are consistent across the forecast notebook, saved CSV, leadership memo and presentation after rounding. The scenario definitions and headline values align.

The saved CSV uses presentation-oriented column names, while the notebook export code uses machine-oriented names. The values are the same, but the file is not reproduced byte-for-byte by the displayed export cell without an additional rename step.

The repository does not fully pin the software environment. Reproduction requires a compatible Python environment with DuckDB, pandas, Plotly and Jupyter. This affects operational reproducibility but not the conceptual forecast logic.

11. Forecast quality assessment

11.1 Strengths

- Clear separation between consumption revenue and cash collections.
- Point-in-time AR reconstruction avoids look-ahead bias.
- Pipeline is separated from the existing customer base.
- Overlapping Vertex opportunities are not naively summed.
- Large customers receive individual treatment.
- Growth caps prevent unstable short-term ratios from exploding.
- The model is explainable and auditable at customer, product and opportunity level.
- Scenario outputs are internally coherent and easy to communicate to leadership.

11.2 Main weaknesses

- Base is a selected scenario, not a statistically calibrated expected value.
- Some pricing and billing-frequency rules are simplified rather than fully contract-aware.
- Payment timing is modelled more strongly than probability of collection.
- The three-month horizon is highly sensitive to a small number of pipeline deals.
- Several business mappings are inferred because CRM and billing identifiers are not fully reconciled.
- August organic results contain more extrapolation risk than June.
- The Downside isolates commercial weakness but does not represent a full liquidity stress.

11.3 Overall judgement

The model is suitable as a transparent short-horizon planning forecast and interview case solution. It supports management discussion by showing the difference between recurring organic performance, existing receivable collections and conditional deal-driven cash.

It should not be interpreted as a precise probabilistic forecast. The most decision-relevant uncertainty is not ordinary model error; it is the commercial resolution and timing of a small number of large opportunities, particularly Vertex.

12. How leadership should use the forecast

4. Use Base as the current planning scenario, not as a guaranteed or mathematically probability-weighted outcome.
5. Track organic consumption separately from CRM pipeline contribution.
6. Track recurring/AR cash separately from annual prepayments.
7. Treat July cash as highly conditional on deal timing and scope.
8. Refresh the model immediately when June actual usage, major invoice payments or CRM status changes become available.

9. For liquidity planning, supplement the current Downside with a separate collections stress case if needed.

13. What the forecast does not claim

- It does not claim that every Base pipeline opportunity has a 50% or 90% statistical probability.
- It does not claim that all future invoices will be collected without loss.
- It does not claim that the historical current-customer cohort equals total company history.
- It does not claim that annual-prepayment timing is recurring monthly operating cash.
- It does not claim that the CRM-to-billing account mapping is fully resolved.
- It does not claim that August has the same confidence as June.
- It does not replace a contract-level Finance review or Sales validation of major deals.

Appendix A. Compact methodology summary

Consumption revenue = forecast usage of current active customers, priced at applicable list prices and discounts, plus selected CRM opportunity consumption.

Organic usage = May customer-product quantity compounded by capped March-May coefficients. Two large customers use individual coefficients; all others use product-level coefficients.

Cash collections = open AR as of 6 June 2026 + payment-lagged future postpaid usage + scenario-selected annual prepayments.

Downside, Base and Upside differ through organic coefficients, specific customer exclusions and selected CRM opportunities. Existing AR is unchanged across scenarios.

Appendix B. Terminology

Term	Meaning in this model
Organic base	Forecast contribution from customers currently active in billing.
Pipeline	Incremental contribution from selected CRM opportunities.
Consumption revenue	Value of services consumed in the service month.
Cash collections	Cash expected to be received in the calendar month.
Open AR	Invoices open as of the cut-off date after point-in-time reconstruction.
Payment lag	Months between service month and paid month among historical paid invoices.
Annual prepayment	Upfront cash based on committed spend and contract term.
Scenario assumption	A deliberate management case, not necessarily a statistical probability.